

Recovery–Decorrelation Uncertainty at the Coherent Möbius Gate: Weighted Phase Frames, Sharp Fourier Frontiers, and Equality–Slice Rigidity

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July 2026

Abstract

TPC–39 constructs a minimally sampled additive phase bank that recovers the equality alias exactly. Its positive square energy nevertheless contains the original equality energy, and its determinant phases are constant across physical rows at a fixed alias. A natural continuation is to add row phases and try to decorrelate the surviving cross–row Möbius form. We prove a sharp obstruction to every such coefficient–blind positive–energy scheme.

Let $\Phi = (\phi_{t\alpha}) \in \mathbb{C}^{m \times R}$ have unimodular entries, let $w_t > 0$ with $\sum_t w_t = 1$, and suppose coefficients a_t recover the coherent row sum universally:

$$\sum_{t=1}^m a_t \phi_{t\alpha} = 1 \quad (1 \leq \alpha \leq R).$$

Define the recovery amplification and normalized row Gram by

$$\mathcal{A} = \sum_t \frac{|a_t|^2}{w_t}, \quad G = \Phi^* \text{diag}(w) \Phi.$$

Then the exact Loewner domination

$$\boxed{G \succeq \mathcal{A}^{-1} \mathbf{1} \mathbf{1}^*}$$

holds. Consequently

$$\mathcal{A} \lambda_{\max}(G) \geq R, \quad \sum_{\alpha \neq \beta} \text{Re } G_{\alpha\beta} \geq R^2/\mathcal{A} - R.$$

The first inequality is a recovery–decorrelation uncertainty law. It is sharp at every integer point of its Pareto frontier: taking any r rows of the R -point Fourier matrix, including the constant row, gives $\mathcal{A} = r$ and $\lambda_{\max}(G) = R/r$.

The equality cases are rigid. Unit amplification $\mathcal{A} = 1$ is possible only when all phase columns are identical, in which case $G = \mathbf{1} \mathbf{1}^*$ and no row decorrelation occurs. Conversely, complete row orthogonality $G = I_R$ forces $\mathcal{A} \geq R$, with equality for the full Fourier bank. We also prove a pseudoinverse formula for the optimal amplification, an L^2 -stable approximate–recovery version, and exact diagonal invariance for Hilbert–valued row atoms.

Tensoring an arbitrary row bank with the $q = L + 1$ alias bank of TPC–39 does not alter this tradeoff. The alias factor has unit amplification; on the equality slice the tensor Gram is exactly G . More general coupled row–alias phases obey the same lower bound after restriction to the zero–alias slice. Thus determinant twisting leaves maximal row coherence,

while Fourier row decorrelation pays back exactly its gain in recovery amplification and leaves the atomic diagonal unchanged.

At the prime–Möbius endpoint this closes a method class, not the arithmetic gate. It proves that universal phase geometry and a positive square–energy estimate cannot by themselves remove the fixed–alias cross–row form. It does not prove a two– or four–Möbius correlation estimate, a parity breach, a Hardy–Littlewood asymptotic, or the twin–prime conjecture. Any continuation inside this framework must break at least one of its hypotheses; concrete options include using the actual arithmetic coefficients, retaining a signed nonpositive functional, or proving an equivalent coefficient–specific dispersion theorem.

Keywords: Möbius correlation; exact recovery; phase frame; coherent mode; Fourier bank; CRT alias; parity boundary.

MSC 2020: Primary 42C15, 15A18; Secondary 11L07, 11N35, 42A16.

1 Introduction

The last three layers of the TPC program separate three different obstructions at an orbit–sliced prime–Möbius energy gate. TPC–37 closes the zero coordinate faces and the proper multiplicative faces, but the full equality face retains the original physical vector with coefficient $1 - o(1)$. TPC–38 computes the target–synthesis spectrum and shows that multiplicative completion produces a coherent joint–alias bucket rather than the single equality column. TPC–39 constructs the missing additive quotient projector: exactly $L + 1$ twists recover the zero alias, and no smaller arbitrary phase bank does so universally [14–16].

That algebraic recovery does not estimate the output. Its minimal positive square energy has the folded Parseval form

$$\|Z_0\|^2 + \sum_{s=1}^L \|Z_s + Z_{s-q}\|^2, \quad q = L + 1. \quad (1)$$

In particular, $\|Z_0\|^2$ is present as a positive summand. At the terminal layer, the alias phase is also independent of the physical row. It therefore does not oscillate the fixed–alias cross–row quadratic form, whose coefficients contain genuine two– and four–Möbius correlations.

This suggests a second phase coordinate. One could assign phases to the physical rows, average the resulting energies, and hope to replace a coherent row Gram by an almost diagonal one. The difficulty is that the same bank must still recover the original coherent row sum. The present paper determines the exact tradeoff between those two requirements.

1.1 Main finite theorem

Let R be the number of active rows, let Φ be a unimodular $m \times R$ phase matrix, and let w be a positive probability vector on its measurements. If a recovers the coherent sum, define

$$\mathcal{A}(a, w) = \sum_{t=1}^m \frac{|a_t|^2}{w_t}, \quad G(\Phi, w) = \Phi^* \text{diag}(w) \Phi. \quad (2)$$

Our core result is the matrix inequality

$$\boxed{G(\Phi, w) \succeq \frac{1}{\mathcal{A}(a, w)} \mathbf{1}\mathbf{1}^*}. \quad (3)$$

It is stronger than an operator–norm estimate: every positive bank energy that supports universal recovery already contains a fixed fraction of the coherent target energy. Since G has unit diagonal, one gets the sharp alternatives

$$\mathcal{A} = 1 \implies G = \mathbf{1}\mathbf{1}^*, \quad G = I_R \implies \mathcal{A} \geq R. \quad (4)$$

Partial Fourier banks attain every integer point

$$(\mathcal{A}, \lambda_{\max}(G)) = (r, R/r), \quad 1 \leq r \leq R, \quad (5)$$

so the product law is not an artifact of the proof.

1.2 Why this is the correct next layer

The theorem does not claim that the physical Möbius vector is a worst case vector for G . It rules out a narrower and precisely declared method class: coefficient-blind, equality-preserving phase banks that are estimated through a positive weighted square energy. Within that class, row decorrelation is paid for by exactly the recovery amplification, while the atomic diagonal is invariant. A large sieve or tight-frame statement about the phase geometry cannot alter this conservation law.

This is useful even though it is an obstruction. Before this paper, adding row signs or a second Fourier coordinate remained a plausible formal escape from the fixed-alias cross-row term in TPC-39. The sharp Loewner theorem closes that escape without making any conjecture about Möbius signs. What remains is necessarily coefficient specific: one must show that the actual arithmetic vector avoids the coherent mode, or estimate a signed bilinear functional before replacing it by positive energy.

1.3 Guide to the paper

[Section 2](#) records the inherited alias and endpoint ledgers. The weighted phase formalism, Loewner theorem, equality cases, and sharp Fourier families occupy the next four sections. We then tensor the result with quotient tomography, treat coupled row-alias phases, and specialize the conservation law to the physical endpoint. A comparison with multiplicative superwindows and with the nearest frame, large-sieve, and Möbius results keeps the scope explicit. The final sections state the coefficient-specific next gate and the exact computational certificate.

1.4 Claim boundary

All phase-frame and matrix results in this paper are unconditional. The physical consequence is a no-go for the stated universal positive-energy method class. We do not prove that the actual cross-row energy is large, that every phase method fails, or that signed coefficient-specific dispersion is impossible. In particular, no affine Chowla estimate, parity-breaking estimate, Hardy-Littlewood formula, or twin-prime theorem is asserted.

2 Inherited equality slice and endpoint ledger

We restate only the part of TPC-39 needed to connect the finite phase theorem to the physical packet. All estimates below that refer to Möbius cancellation remain explicitly conditional.

2.1 Rows and aliases

On one retained packet,

$$QJ \asymp X, \quad Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}. \quad (6)$$

Let $\mathcal{A}$ be its active row set, of cardinality

$$R_{\text{row}} := |\mathcal{A}| \ll X^{o(1)}Q. \quad (7)$$

The main finite theorems use an arbitrary integer R ; physical specializations are stated for any selected R -row subpacket. When a dense packet with $R = X^{o(1)}Q$ is invoked, that density is written as an additional hypothesis rather than inferred from (7).

Choose the three orbit-scale primes of TPC-39 and put

$$M = q_1 q_2 q_3 = X^{399/400+o(1)}. \quad (8)$$

Every completed determinant lies in $|F| \leq B$, where $B \asymp X$. Define

$$L = \left\lfloor \frac{B}{M} \right\rfloor, \quad q = L + 1, \quad K_B = 2L + 1. \quad (9)$$

Then

$$q \asymp K_B \asymp X^{1/400+o(1)}, \quad qM > B. \quad (10)$$

For each row α , alias $-L \leq k \leq L$, and any remaining physical output index, gather the actual row coefficient, mask, Möbius atom, and cutoff into a Hilbert vector

$$x_{\alpha,k} \in \mathcal{H}. \quad (11)$$

The raw alias column and equality output are

$$\mathcal{Z}_k = \sum_{\alpha \in \mathcal{A}} x_{\alpha,k}, \quad \mathcal{R}_0 = \mathcal{Z}_0. \quad (12)$$

The finite theorems treat the $x_{\alpha,k}$ as arbitrary. This is exactly what makes their later physical conclusion coefficient blind.

2.2 Minimal alias tomography

Let $\omega = \exp(2\pi i/q)$. The TPC-39 measurements are

$$\mathcal{Y}_\nu = \sum_{k=-L}^L \omega^{\nu k} \mathcal{Z}_k, \quad 0 \leq \nu < q. \quad (13)$$

They recover equality with unit normalized amplification:

$$\mathcal{Z}_0 = \frac{1}{q} \sum_{\nu=0}^{q-1} \mathcal{Y}_\nu. \quad (14)$$

Their normalized Gram in the alias coordinate is

$$(H_q)_{k\ell} = \mathbf{1}_{k \equiv \ell \pmod{q}}. \quad (15)$$

Because $q = L + 1$, its scalar spectrum is

$$2 \quad (L \text{ times}), \quad 1 \quad (1 \text{ time}), \quad 0 \quad (L \text{ times}). \quad (16)$$

The eigenvalue one belongs to the isolated zero alias. The eigenvalues two and zero belong to folded nuisance pairs; they must not be confused with the equality-slice recovery constant.

2.3 Energy budget and the surviving term

Put

$$B_0 = Q^2 J. \quad (17)$$

The inherited target is

$$\|\mathcal{R}_0\|^2 \ll X^{o(1)} \frac{Q^3}{J}, \quad (18)$$

and the exact exponent identity is

$$B_0 K_B = X^{o(1)} \frac{Q^3}{J}. \quad (19)$$

TPC-39 closes the complete terminal atomic diagonal at this scale. The unproved part is off diagonal. At fixed alias it contains cross-row terms, and between the folded aliases it contains a gap of size $qM \in (B, B+M]$. The determinant phase $\omega^{\nu k}$ is constant in α , so no row cancellation follows from (13).

The purpose of the present paper is to decide whether an additional coefficient-blind row phase can repair this last defect while preserving (14) at low amplification. It cannot; the exact obstruction is finite dimensional and independent of any unproved statement about Möbius correlations.

3 Weighted phase banks and conjugation conventions

Fix positive integers m and R . A phase bank is a matrix

$$\Phi = (\phi_{j\alpha})_{1 \leq j \leq m, 1 \leq \alpha \leq R} \in \mathbb{C}^{m \times R}, \quad |\phi_{j\alpha}| = 1. \quad (20)$$

Its rows are measurements and its columns are the R signatures to be recovered with the same unit gain. Let

$$w_j > 0, \quad \sum_{j=1}^m w_j = 1, \quad W := \text{diag}(w_1, \dots, w_m). \quad (21)$$

We use the following complex conventions throughout. For a column x , x^T denotes ordinary transpose, $x^* = \bar{x}^T$ denotes conjugate transpose, and

$$\langle x, y \rangle = x^* y \quad (22)$$

is conjugate linear in the first variable and linear in the second. The recovery coefficients form a column $a \in \mathbb{C}^m$, but the recovery rule uses ordinary transpose:

$$a^T \Phi = \mathbf{1}_R^T. \quad (23)$$

Thus no conjugate is silently inserted into the measurement rule. Taking the conjugate transpose of (23) gives the equivalent equation

$$\Phi^* \bar{a} = \mathbf{1}_R. \quad (24)$$

Define the weighted normalized amplification and weighted Gram matrix by

$$A_W(a) := a^* W^{-1} a = \sum_{j=1}^m \frac{|a_j|^2}{w_j}, \quad (25)$$

$$G_W(\Phi) := \Phi^* W \Phi. \quad (26)$$

When $w_j = 1/m$, these reduce to

$$A_W(a) = m\|a\|_2^2, \quad G_W(\Phi) = \frac{1}{m}\Phi^*\Phi. \quad (27)$$

We normally abbreviate them to A and G . The physical-interface sections use the typographic form $\mathcal{A}$ for the same amplification.

It is useful to absorb the weights into

$$D := W^{1/2}\Phi, \quad b := W^{-1/2}a, \quad z := \bar{b} = W^{-1/2}\bar{a}. \quad (28)$$

Then

$$G = D^*D, \quad A = \|b\|_2^2 = \|z\|_2^2, \quad D^*z = \mathbf{1}_R. \quad (29)$$

The last identity is the adjoint form of $b^T D = \mathbf{1}_R^T$.

Because every phase has modulus one and the weights sum to one,

$$G \succeq 0, \quad G_{\alpha\alpha} = 1, \quad \text{tr } G = R. \quad (30)$$

In particular,

$$\lambda_{\max}(G) \geq 1, \quad (31)$$

with equality if and only if $G = I_R$. The latter assertion follows because all R nonnegative eigenvalues must then equal their common upper bound 1.

4 The recovery–coherence uncertainty theorem

The weighted formulation gives a matrix inequality, not only a scalar uncertainty product.

Theorem 4.1 (Exact weighted Loewner decomposition). *Assume (23), and let A , G , D , and b be as in section 3. Put*

$$u := \frac{\bar{b}}{\sqrt{A}}. \quad (32)$$

Then $u^*u = 1$ and

$$\boxed{G - \frac{1}{A}\mathbf{1}_R\mathbf{1}_R^* = D^*(I_m - uu^*)D \succeq 0.} \quad (33)$$

Consequently,

$$\boxed{A\lambda_{\max}(G) \geq R.} \quad (34)$$

Proof. By (29),

$$u^*D = \frac{b^T D}{\sqrt{A}} = \frac{1}{\sqrt{A}}\mathbf{1}_R^T.$$

Taking adjoints gives $D^*u = A^{-1/2}\mathbf{1}_R$, and therefore

$$D^*uu^*D = \frac{1}{A}\mathbf{1}_R\mathbf{1}_R^*.$$

Subtracting this identity from $D^*D = G$ proves (33). Monotonicity of the largest eigenvalue in the Loewner order gives

$$\lambda_{\max}(G) \geq \lambda_{\max}\left(\frac{1}{A}\mathbf{1}_R\mathbf{1}_R^*\right) = \frac{R}{A},$$

which is (34). □

Equivalently, for every $x \in \mathbb{C}^R$, weighted Cauchy–Schwarz gives

$$|\mathbf{1}_R^T x|^2 = |a^T \Phi x|^2 \leq A x^* G x. \quad (35)$$

The factorization in (33) records all such inequalities at once.

4.1 Minimum amplification and the pseudoinverse

The exact recovery system is feasible if and only if $\mathbf{1}_R \in \text{Ran } D^* = \text{Ran } G$. Write $G^\dagger$ for the Moore–Penrose pseudoinverse.

Proposition 4.2 (Minimum-amplification recovery). *If exact recovery is feasible, then its minimum weighted amplification is*

$$\boxed{A_{\min} = \mathbf{1}_R^* G^\dagger \mathbf{1}_R.} \quad (36)$$

The unique minimum-norm normalized variable and corresponding recovery coefficient are

$$z_{\min} = DG^\dagger \mathbf{1}_R, \quad (37)$$

$$a_{\min} = W\overline{\Phi} \overline{G^\dagger \mathbf{1}_R}. \quad (38)$$

Every solution has the form

$$W^{-1/2}\overline{a} = z_{\min} + h, \quad h \in \ker D^*, \quad (39)$$

and satisfies

$$A = A_{\min} + \|h\|_2^2. \quad (40)$$

Proof. The minimum-norm solution of $D^*z = \mathbf{1}_R$ is $z_{\min} = (D^*)^\dagger \mathbf{1}_R = DG^\dagger \mathbf{1}_R$. Since G is Hermitian and $\mathbf{1}_R \in \text{Ran } G$,

$$\|z_{\min}\|_2^2 = \mathbf{1}_R^* G^\dagger G G^\dagger \mathbf{1}_R = \mathbf{1}_R^* G^\dagger \mathbf{1}_R.$$

The relation between z and a in (28) yields (38). Finally, $z_{\min} \in \text{Ran } D$ is orthogonal to $\ker D^*$, which proves (40). $\square$

4.2 Equality and endpoint rigidity

Proposition 4.3 (Equality in the uncertainty product). *Let $\lambda = \lambda_{\max}(G)$. Under exact recovery, equality in (34) holds if and only if*

$$G\mathbf{1}_R = \lambda\mathbf{1}_R, \quad W^{-1/2}\overline{a} = \frac{1}{\lambda}D\mathbf{1}_R. \quad (41)$$

Equivalently,

$$a = \frac{1}{\lambda}W\overline{\Phi}\mathbf{1}_R. \quad (42)$$

Thus an equality recovery is the minimum-amplification recovery for its bank, and $A = R/\lambda$.

Proof. The scalar proof of (34) is

$$R = \|D^*z\|_2^2 = z^*DD^*z \leq \lambda\|z\|_2^2 = \lambda A.$$

Equality holds precisely when z lies in the top eigenspace of DD^* . Since $D^*z = \mathbf{1}_R$, the eigenvector equation is

$$D\mathbf{1}_R = DD^*z = \lambda z.$$

Applying D^* also gives $G\mathbf{1}_R = \lambda\mathbf{1}_R$. These implications are reversible, and translation from z to a proves the result. $\square$

Proposition 4.4 (Unit-amplification rigidity). *Every exact recovery has $A \geq 1$. Moreover, $A = 1$ if and only if there is a phase column $v \in \mathbb{C}^m$, $|v_j| = 1$, such that*

$$\Phi = v \mathbf{1}_R^T, \quad a_j = w_j \bar{v}_j \quad (1 \leq j \leq m). \quad (43)$$

In this case $G = \mathbf{1}_R \mathbf{1}_R^$ and $\lambda_{\max}(G) = R$.*

Proof. For every column ϕ_α of Φ ,

$$1 = |a^T \phi_\alpha|^2 \leq \left(\sum_j \frac{|a_j|^2}{w_j} \right) \left(\sum_j w_j |\phi_{j\alpha}|^2 \right) = A.$$

If $A = 1$, equality holds for every column. In the normalized variables, each D -column has norm one and has inner product one with the unit vector $z = \bar{b}$. Hence every D -column equals z , so all columns of Φ are the same phase column v . The equality vector is $z = W^{1/2}v$, which is equivalent to $a_j = w_j \bar{v}_j$. The converse is immediate. $\square$

At the opposite endpoint, complete Gram decorrelation has a precisely quantified recovery cost.

Proposition 4.5 (Cost of an orthogonal Gram). *If $G = I_R$, then exact recovery is feasible and every exact recovery can be written as*

$$W^{-1/2} \bar{a} = D \mathbf{1}_R + h, \quad h \in \ker D^*. \quad (44)$$

Its amplification is

$$A = R + \|h\|_2^2 \geq R. \quad (45)$$

Equality holds exactly for

$$a = W \bar{\Phi} \mathbf{1}_R. \quad (46)$$

In particular, the simultaneous conditions $A = 1$ and $G = I_R$ are possible only when $R = 1$.

Proof. When $D^*D = I_R$, the vector $D \mathbf{1}_R$ solves $D^*z = \mathbf{1}_R$, has squared norm R , and is orthogonal to $\ker D^*$. The asserted formulas follow from the orthogonal decomposition of all solutions. $\square$

5 Fourier extremizers and the sharp Pareto family

The product lower bound is attained at every integer point between its two endpoints. The relevant cyclic bank uses rows of the R -point Fourier matrix; replacing the R -point phases by a shorter modulus r gives a different construction and introduces an integrality defect.

Let

$$\zeta_R := \exp(2\pi i/R). \quad (47)$$

Theorem 5.1 (Partial-DFT extremizers). *Fix $1 \leq r \leq R$, choose any $T \subset \mathbb{Z}/R\mathbb{Z}$ with $|T| = r$ and $0 \in T$, and let $s \geq 1$. Put $m = rs$, index the rows by $(t, \ell) \in T \times \{1, \dots, s\}$, and the columns by $\alpha \in \mathbb{Z}/R\mathbb{Z}$. Define*

$$\Phi_{(t, \ell), \alpha} = \zeta_R^{t\alpha}, \quad w_{(t, \ell)} = \frac{1}{rs}, \quad a_{(t, \ell)} = \frac{1}{s} \mathbf{1}_{t=0}. \quad (48)$$

Then

$$a^T \Phi = \mathbf{1}_R^T, \quad A = r, \quad (49)$$

and the Gram spectrum is

$$\text{spec}(G) = \left\{ \frac{R}{r} \text{ with multiplicity } r, \quad 0 \text{ with multiplicity } R - r \right\}. \quad (50)$$

In particular,

$$A\lambda_{\max}(G) = r \frac{R}{r} = R. \quad (51)$$

Hence every pair

$$(A, \lambda_{\max}(G)) = \left(r, \frac{R}{r} \right), \quad 1 \leq r \leq R, \quad (52)$$

lies on the sharp Pareto boundary.

Proof. The only rows used by the recovery coefficient are the s copies of the constant Fourier row. Their coefficients sum to one, proving exact recovery. Moreover,

$$A = rs \cdot s \left| \frac{1}{s} \right|^2 = r.$$

The distinct rows selected from the unnormalized R -point Fourier matrix are mutually orthogonal and each has squared norm R . Equal replication therefore gives

$$G = \frac{1}{r} \sum_{t \in T} v_t^* v_t, \quad (v_t)_\alpha = \zeta_R^{t\alpha},$$

whose r nonzero eigenvalues are all R/r . Finally, the constant column vector is an R/r -eigenvector because the zero Fourier row is present. The Pareto assertion follows from [theorem 4.1](#): a strict improvement in one coordinate without worsening the other would make the product smaller than R . $\square$

The endpoint $r = 1$ is the completely coherent, unit-amplification bank; $r = R$ is the full DFT with $G = I_R$ and $A = R$.

5.1 A continuous weighted boundary

Assume $R \geq 2$. The weighted model also realizes every real point of the hyperbola between the endpoints. For $1 < A_0 \leq R$, take the full R -point Fourier matrix, put

$$w_0 = \frac{1}{A_0}, \quad w_t = \frac{A_0 - 1}{A_0(R - 1)} \quad (1 \leq t < R), \quad a = e_0, \quad (53)$$

where e_0 selects the constant Fourier row.

Corollary 5.2 (Continuous weighted Pareto boundary). *The bank in (53) satisfies*

$$A = A_0, \quad \lambda_{\max}(G) = \frac{R}{A_0}, \quad A\lambda_{\max}(G) = R. \quad (54)$$

The missing endpoint $A_0 = 1$, excluded from this R -row formula by the strict positivity of all w_t , is realized by the one-row constant bank.

Proof. Writing F_R for the unnormalized Fourier matrix, one has $G = F_R^* W F_R$. Since $R^{-1/2} F_R$ is unitary, the eigenvalues of G are Rw_t . For $A_0 \leq R$, $w_0 \geq w_t$ for $t \neq 0$, so the largest eigenvalue is R/A_0 . Also $A = |a_0|^2/w_0 = A_0$. $\square$

For uniform rows, rational points in this continuous family can equally be implemented by repeating Fourier rows in the corresponding proportions.

5.2 The short-cycle bucket defect

There is a tempting but inequivalent construction. Take $m = r$, use phases modulo r , and index the R columns consecutively:

$$\Phi_{t,\alpha}^{\text{short}} = \exp(2\pi i t \alpha / r), \quad 0 \leq t < r, \quad 0 \leq \alpha < R, \quad (55)$$

with uniform row weights and recovery $a = e_0$. Then $A = r$, but

$$G_{\alpha\beta}^{\text{short}} = \mathbf{1}_{\alpha \equiv \beta \pmod{r}}. \quad (56)$$

Its nonzero eigenvalues are the sizes of the r residue buckets, hence

$$\lambda_{\max}(G^{\text{short}}) = \left\lceil \frac{R}{r} \right\rceil. \quad (57)$$

Therefore this balanced short-cycle bank lies on the sharp Pareto boundary if and only if $r \mid R$. If $r \nmid R$, the partial R -point DFT bank of [theorem 5.1](#) has the same amplification r and the strictly smaller coherence radius R/r . Thus balance within the short-cycle subclass does not by itself prove global Pareto optimality.

6 Stability under approximate recovery

The exact Loewner argument has a direct biased version. Let $a \neq 0$ be arbitrary and define its gain vector $g \in \mathbb{C}^R$ by

$$g^T := a^T \Phi. \quad (58)$$

Thus exact unit-gain recovery is the special case $g = \mathbf{1}_R$. Retain $A = a^* W^{-1} a$, $D = W^{1/2} \Phi$, $b = W^{-1/2} a$, and $u = \bar{b} / \sqrt{A}$.

Theorem 6.1 (Approximate weighted Loewner inequality). *For every nonzero recovery coefficient a ,*

$$\boxed{G - \frac{1}{A} \bar{g} g^T = D^*(I_m - uu^*)D \succeq 0.} \quad (59)$$

Consequently,

$$A\lambda_{\max}(G) \geq \|g\|_2^2. \quad (60)$$

If, for some $0 \leq \varepsilon \leq 1$,

$$\|g - \mathbf{1}_R\|_2 \leq \varepsilon \sqrt{R}, \quad (61)$$

then

$$\boxed{A\lambda_{\max}(G) \geq R(1 - \varepsilon)^2.} \quad (62)$$

Proof. The definition of g gives

$$u^* D = \frac{b^T D}{\sqrt{A}} = \frac{1}{\sqrt{A}} g^T, \quad D^* u = \frac{1}{\sqrt{A}} \bar{g}.$$

This proves the factorization in (59). The rank-one matrix $A^{-1} \bar{g} g^T$ has largest eigenvalue $A^{-1} \|g\|_2^2$, proving (60). Finally, the reverse triangle inequality and (61) give

$$\|g\|_2 \geq \|\mathbf{1}_R\|_2 - \|g - \mathbf{1}_R\|_2 \geq (1 - \varepsilon) \sqrt{R}.$$

Squaring completes the proof. $\square$

Corollary 6.2 (Coordinatewise error). *If*

$$\max_{1 \leq \alpha \leq R} |g_\alpha - 1| \leq \varepsilon \quad (0 \leq \varepsilon \leq 1), \quad (63)$$

then (62) holds.

Proof. The coordinatewise assumption implies $\|g - \mathbf{1}_R\|_2 \leq \varepsilon \sqrt{R}$. $\square$

The norm and normalization in (61) are essential. Without a quantitative condition that forces $\|g\|_2$ to remain close to $\sqrt{R}$, no lower bound with the factor $(1 - \varepsilon)^2$ follows. Under the same error hypothesis, but for unrestricted $\varepsilon \geq 0$, the proof yields the always-valid form

$$\mathcal{A} \lambda_{\max}(G) \geq R(1 - \varepsilon)_+^2, \quad (x)_+ := \max\{x, 0\}. \quad (64)$$

7 Tensoring row phases with minimal alias tomography

We now combine the row theorem with the $q = L + 1$ quotient bank from section 2. The tensor calculation separates the equality slice from the folded nuisance spectrum and shows that alias tomography supplies no hidden row decorrelation.

7.1 The separable product bank

Let $\Phi = (\phi_{t\alpha})$ be a weighted row phase bank with weights w_t , recovery coefficients a_t , amplification $\mathcal{A}$, and Gram G . Define

$$\mathcal{Y}_{t,\nu} := \sum_{\alpha=1}^R \sum_{k=-L}^L \phi_{t\alpha} \omega^{\nu k} x_{\alpha,k}, \quad 0 \leq \nu < q. \quad (65)$$

Give the pair (t, ν) weight w_t/q and recovery coefficient

$$\tilde{a}_{t,\nu} = \frac{a_t}{q}. \quad (66)$$

Theorem 7.1 (Tensor recovery and amplification conservation). *The product bank recovers the coherent equality slice:*

$$\sum_{t,\nu} \tilde{a}_{t,\nu} \mathcal{Y}_{t,\nu} = \sum_{\alpha=1}^R x_{\alpha,0}. \quad (67)$$

Its normalized recovery amplification is exactly

$$\tilde{\mathcal{A}} = \mathcal{A}, \quad (68)$$

and its complete row-alias Gram is

$$\tilde{G} = G \otimes H_q, \quad (69)$$

where H_q is given by (15). On the zero-alias coordinate block,

$$\tilde{G}|_{k=\ell=0} = G. \quad (70)$$

Consequently

$$\boxed{\mathcal{A} \lambda_{\max}(\tilde{G}|_{k=\ell=0}) \geq R.} \quad (71)$$

Proof. The coefficient of $x_{\alpha,k}$ in the left side of (67) is

$$\left(\sum_t a_t \phi_{t\alpha} \right) \left(\frac{1}{q} \sum_{\nu=0}^{q-1} \omega^{\nu k} \right) = \mathbf{1}_{k=0}.$$

Moreover,

$$\sum_{t,\nu} \frac{|\tilde{a}_{t,\nu}|^2}{w_t/q} = \sum_t \frac{|a_t|^2}{w_t} = \mathcal{A}.$$

The Gram factorization follows by separating the two phase sums. Since zero is a singleton residue in $[-L, L]$, $(H_q)_{00} = 1$, proving (70). Apply the recovery–coherence theorem on that block. $\square$

The full tensor operator has a nuisance factor two. If $\lambda_1, \dots, \lambda_R$ are the eigenvalues of G , then for each nonzero λ_j the corresponding nonzero tensor eigenvalues are

$$2\lambda_j \quad (L \text{ copies for each } j), \quad \lambda_j \quad (1 \text{ copy for each } j). \quad (72)$$

Zeros arise from the folded alias kernel and, when G is singular, also from the outer row bank. Thus the full operator norm is $2\|G\|_{\text{op}}$ when $L \geq 1$, but the relevant equality–slice coherence is $\|G\|_{\text{op}}$. We do not insert the irrelevant factor two into the sharp recovery tradeoff.

Corollary 7.2 (Partial Fourier tensor spectrum). *For an r -row partial Fourier extremizer, the product bank has*

$$\tilde{\mathcal{A}} = r, \quad (73)$$

and nonzero scalar spectrum

$$\frac{2R}{r} \quad (rL \text{ times}), \quad \frac{R}{r} \quad (r \text{ times}). \quad (74)$$

The equality–slice eigenvalue is R/r , and its product with the recovery amplification is exactly R .

7.2 Coupled phases do not evade the zero slice

Separable phases are not essential. Let s index an arbitrary bank with positive weights w_s , and let

$$\theta_{s,\alpha,k} \in \mathbb{T} \quad (75)$$

depend jointly on row and alias. Suppose coefficients a_s recover the coherent equality slice universally:

$$\sum_s a_s \theta_{s,\alpha,k} = \mathbf{1}_{k=0} \quad (1 \leq \alpha \leq R, \quad -L \leq k \leq L). \quad (76)$$

Put $\mathcal{A} = \sum_s |a_s|^2 / w_s$.

Theorem 7.3 (Equality–slice rigidity for coupled banks). *Let*

$$G_{\alpha\beta}^{(0)} := \sum_s w_s \overline{\theta_{s,\alpha,0}} \theta_{s,\beta,0}. \quad (77)$$

Then

$$\boxed{G^{(0)} \succeq \mathcal{A}^{-1} \mathbf{1}\mathbf{1}^*, \quad \mathcal{A} \lambda_{\max}(G^{(0)}) \geq R.} \quad (78)$$

For arbitrary Hilbert atoms,

$$\left\| \sum_{\alpha} x_{\alpha,0} \right\|^2 \leq \mathcal{A} \sum_s w_s \left\| \sum_{\alpha,k} \theta_{s,\alpha,k} x_{\alpha,k} \right\|^2. \quad (79)$$

Proof. Restrict (76) to $k = 0$. The phase columns $(\theta_{s,\alpha,0})_s$ form an exact weighted row bank with the same recovery coefficients and amplification. The Loewner theorem gives (78). The Hilbert-valued weighted Cauchy inequality applied to the exact recovery gives (79). $\square$

Thus nonseparable coding, random row signs, and joint row-alias phase design do not bypass the obstruction as long as recovery is universal and the analytic step is a positive average of squared norms. The no-go is obtained by setting every nuisance alias to zero; no behavior away from the target slice can repair it.

7.3 The determinant bank is the unit-amplification endpoint

For the TPC-39 determinant twists,

$$\theta_{\nu,\alpha,k} = \omega^{\nu k} \quad (80)$$

is independent of α . On $k = 0$, every phase column is the same constant vector. Hence

$$\mathcal{A} = 1, \quad G^{(0)} = \mathbf{1}\mathbf{1}^*, \quad \lambda_{\max}(G^{(0)}) = R. \quad (81)$$

The survival of fixed-alias cross-row coherence in TPC-39 is therefore the rigid equality case of the general theorem, not a defect of the chosen proof of folded Parseval.

8 Physical endpoint: what phase geometry can and cannot buy

We now insert the finite tradeoff into the ledger of section 2. Every conclusion in this section is a method statement. It does not assert that the actual Möbius vector saturates a coherent eigenmode.

8.1 The atomic diagonal is invariant

For Hilbert row atoms $x_1, \dots, x_R$, a weighted phase bank has energy

$$\begin{aligned} \mathcal{E}_\Phi(x) &:= \sum_t w_t \left\| \sum_\alpha \phi_{t\alpha} x_\alpha \right\|^2 \\ &= \sum_{\alpha,\beta} G_{\alpha\beta} \langle x_\alpha, x_\beta \rangle. \end{aligned} \quad (82)$$

Since $|\phi_{t\alpha}| = 1$ and $\sum_t w_t = 1$,

$$G_{\alpha\alpha} = 1. \quad (83)$$

Therefore the atomic diagonal in (82) is always

$$\boxed{\sum_{\alpha=1}^R \|x_\alpha\|^2}, \quad (84)$$

independently of the number or choice of row phases. Phasing can only redistribute the off-diagonal terms.

If the bank universally recovers the coherent row sum, the Loewner theorem simultaneously gives

$$\mathcal{E}_\Phi(x) \geq \frac{1}{\mathcal{A}} \left\| \sum_\alpha x_\alpha \right\|^2. \quad (85)$$

Thus an upper bound $\mathcal{E}_\Phi(x) \leq \mathcal{B}$ yields only

$$\left\| \sum_\alpha x_\alpha \right\|^2 \leq \mathcal{A}\mathcal{B}. \quad (86)$$

8.2 The diagonal–coherence dilemma

In the TPC–39 terminal packet, the complete atomic diagonal already has the endpoint size

$$\mathcal{D}_{\text{term}} \ll X^{o(1)} B_0 K_B = X^{o(1)} \frac{Q^3}{J}. \quad (87)$$

Normalized row phasing does not reduce this quantity. Consequently, a coefficient–blind argument that bounds the whole bank energy only at the diagonal scale and then invokes (86) must keep

$$\mathcal{A} = X^{o(1)} \quad (88)$$

to retain the endpoint target. But then every R -row subpacket obeys

$$\lambda_{\max}(G) \geq \frac{R}{\mathcal{A}} = R X^{-o(1)}. \quad (89)$$

Nearly the full coherent row mode remains. Complete row orthogonality would instead require $\mathcal{A} \geq R$, multiplying the already sharp diagonal ledger by R .

The partial Fourier extremizers make the conservation literal. Using r Fourier measurements reduces the coherent eigenvalue from R to R/r , but the recovery cost becomes r . Their product is R , and the diagonal in (84) does not move. No choice of r turns the known diagonal estimate into the desired coherent estimate without additional arithmetic information.

8.3 A measurement–dimension lower bound

Every m -measurement row Gram has

$$\text{rank } G \leq m, \quad \text{tr } G = R. \quad (90)$$

Hence

$$\lambda_{\max}(G) \geq \frac{R}{m}. \quad (91)$$

Suppose, only for scale comparison, that a dense active packet has

$$R = X^{o(1)} Q \quad (92)$$

and that one attempts to reuse only $m \asymp K_B = X^{1/400+o(1)}$ joint phase slots. Then

$$\lambda_{\max}(G) \geq X^{266/400-o(1)}. \quad (93)$$

This exponent is a conditional dense–packet specialization of the exact finite rank bound; it is not inferred from the row upper bound alone.

8.4 The scoped no–go

Combining the preceding results gives the following precise conclusion.

Corollary 8.1 (Universal diagonal majorants pay the row dimension). *Suppose an exact weighted phase bank satisfies the coefficient–blind Bessel majorant*

$$\mathcal{E}_{\Phi}(x) \leq C \sum_{\alpha=1}^R \|x_{\alpha}\|^2 \quad (94)$$

for every Hilbert space and every row family. Then

$$\boxed{C \geq \lambda_{\max}(G), \quad \mathcal{AC} \geq R.} \quad (95)$$

Consequently its universal recovered bound is

$$\left\| \sum_{\alpha=1}^R x_{\alpha} \right\|^2 \leq \mathcal{AC} \sum_{\alpha=1}^R \|x_{\alpha}\|^2, \quad (96)$$

and the coefficient-blind constant in this route cannot be smaller than R .

Proof. It suffices to take the Hilbert space to be $\mathbb{C}$. Then (94) is the matrix inequality $G \preceq CI_R$, so $C \geq \lambda_{\max}(G)$. Combine this with $\mathcal{A}\lambda_{\max}(G) \geq R$. The recovered estimate follows from (86). $\square$

Thus a universal diagonal-only majorant cannot close the TPC-39 gate at the same diagonal scale on a polynomially large row packet. Unit or subpolynomial recovery amplification leaves a polynomial coherent mode; full decorrelation pays a polynomial recovery loss while preserving the diagonal.

The corollary says nothing about a proof that uses the actual signs and weights of $x_{\alpha,k}$. A coefficient-specific vector may have a small projection onto the coherent eigenspace even when the operator norm is large. Proving precisely that statement would be genuine arithmetic progress and is not ruled out here.

9 Comparison with the multiplicative superwindow door

TPC-39 isolates a second architecture: enlarge the multiplicative CRT modulus until it exceeds the determinant window. This removes aliases without additive tomography, but it does not create Möbius cancellation. The endpoint scales quantify the granularity cost.

9.1 The orbit-prime granularity gap

The inherited multiplicative coordinates use primes $q_i \asymp J$, and their product is

$$M = q_1 q_2 q_3 = X^{399/400+o(1)}. \quad (97)$$

The smallest no-wrap factor required after this product is only

$$q = L + 1 \asymp \frac{B}{M} = X^{1/400+o(1)}. \quad (98)$$

By contrast, an additional coordinate satisfying the same orbit-prime range has size $p_4 \asymp J = X^{133/400+o(1)}$. Its multiplicative superwindow $M_4 = Mp_4$ obeys

$$\frac{M_4}{B} \asymp \frac{p_4}{q} = X^{132/400+o(1)}. \quad (99)$$

Thus the fourth multiplicative coordinate overshoots the exact no-wrap factor by $X^{33/100+o(1)}$. Additive quotient tomography is far more granular: it implements the effective modulus $qM > B$ using q unimodular measurements.

Equation (99) is specific to the inherited architecture in which every new multiplicative coordinate must lie at the orbit scale. It is not a theorem that every possible composite superwindow has this cost; smaller factors would require a new zero-face, unit-coordinate, and physical synthesis analysis.

9.2 No-wrap is not arithmetic closure

For a squarefree product $R = \prod_{i=1}^s p_i$, the full target face has equality coefficient

$$A_R = \prod_{i=1}^s \left(1 - \frac{1}{p_i - 1}\right). \quad (100)$$

With a fixed number of orbit-scale primes,

$$A_R = 1 - O_s(J^{-1}). \quad (101)$$

Hence a no-wrap modulus removes the alias copies but leaves the original equality vector with asymptotic coefficient one. Its dominant target squared singular value is of order R , as computed in TPC-38 and compared with the alias count in TPC-39. The superwindow therefore trades alias multiplicity for multiplicative synthesis scale; it does not weaken the equality Möbius form.

The additive and multiplicative doors have complementary strengths. Additive tomography reaches the exact no-wrap scale with optimal recovery amplification, but the present theorem shows that a further universal row phase cannot decorrelate for free. A fourth multiplicative coordinate removes aliases directly, but overresolves the window and retains the full equality coefficient. Both routes return to the same coefficient-specific arithmetic boundary.

10 Literature interface and the arithmetic boundary

The phase banks considered here sit at an interface between two theories with very different logical content. Finite-frame and large-sieve inequalities describe the geometry of the measurement map. Results on correlations of multiplicative functions describe cancellation in the measured coefficients. The former can optimize sampling, conditioning, and erasure tolerance, but it cannot supply the latter. This section records the exact scope of the relevant literature so that no deterministic frame statement is mistaken for the twist-stable Möbius estimate left open by TPC-39.

10.1 What finite-frame geometry does prove

Let $v_1, \dots, v_m$ be unit vectors in a d -dimensional complex Hilbert space. The first Welch bound gives, for $m > d$,

$$\max_{r \neq s} |\langle v_r, v_s \rangle|^2 \geq \frac{m - d}{d(m - 1)}. \quad (102)$$

Equality requires the vectors to form an equiangular tight frame [17]. Thus the Welch bound is a lower bound for worst-pair coherence. It is not, by itself, a theorem about a prescribed recovery functional, and it does not say that deleting a measurement preserves exact recovery.

The averaged analogue is the frame potential

$$\text{FP}(v_1, \dots, v_m) := \sum_{r,s=1}^m |\langle v_r, v_s \rangle|^2. \quad (103)$$

For $m \geq d$,

$$\text{FP}(v_1, \dots, v_m) \geq \frac{m^2}{d}, \quad (104)$$

with equality precisely at unit-norm tight frames [1]. This is the appropriate background when a phase bank is optimized by total squared correlation rather than maximum coherence. Tightness minimizes a declared geometric energy; it does not make structured arithmetic coefficients independent.

There is a necessary qualification in the present setting. A minimally sampled phase map on all $2L + 1$ alias coordinates has a nontrivial kernel. Consequently it has no finite condition number as an operator on the full alias space. A statement such as “unit condition number” is valid only after the quotient, range, or recoverable subspace has been specified. The regular polygon bank is an exact Fourier system on its folded quotient; that fact neither removes the folded kernel nor controls the norm of a physical Möbius synthesis.

The analytic large sieve supplies a complementary Bessel bound. If $x_1, \dots, x_R$ are separated modulo one by at least δ , then for a trigonometric polynomial supported on an interval of N integers one has, up to harmless endpoint conventions,

$$\sum_{r=1}^R \left| \sum_n a_n e(nx_r) \right|^2 \ll (N + \delta^{-1}) \sum_n |a_n|^2. \quad (105)$$

This is the separated–frequency form established sharply by Montgomery and Vaughan [10]. It bounds the norm of an analysis operator by the input ℓ^2 -mass. It does not decrease that mass, and it does not force cancellation between two Möbius atoms. In the TPC–39 folded Parseval identity the equality column $\|\mathcal{Z}_0\|^2$ is already a positive summand. No application of (105) can erase this summand without an additional arithmetic estimate.

The division of labor is therefore exact:

$$\boxed{\text{frame geometry does not imply arithmetic decorrelation.}} \quad (106)$$

10.2 Nearby Möbius theorems and their quantifiers

Several deep results come close to one face of the physical gate, but each changes at least one essential quantifier.

First, Klurman et al. [6] prove small variance for sums of a bounded multiplicative function in residue classes modulo q , for almost all $q = o(x)$, with stronger exceptional–set conclusions in special modulus families and an every–modulus conclusion under their stated GRH hypothesis. This is the closest existing theorem to the large–modulus, short–progression geometry of the terminal packet. It is nevertheless a scalar residue–class variance theorem. It does not include the prescribed CRT modulus, the physical row mask, coherent row synthesis, or the affine four–Möbius form. At the endpoint $K = x/M$, its schematic error

$$\varepsilon \varphi(M) K^2 \quad (107)$$

reaches the scalar diagonal scale MK only if $\varepsilon \ll K^{-1} x^{o(1)}$. The available Möbius savings are logarithmic rather than this fixed power. GRH changes the exceptional modulus quantifier, not this missing factor.

Second, Teräväinen and Walker [13] prove a logarithmically averaged two–point Elliott theorem on convex Bohr sets. Their additive–character lemma gives cancellation for two fixed affine forms twisted by a fixed linear phase. This is especially relevant conceptually because the terminal two–Möbius term can also be displayed as an affine correlation with a phase. In the present problem, however, the affine coefficients, shifts, modulus, and rational phase denominator all move with X ; the desired quantity is a natural square energy with row weights. Moreover, their higher–point statement generally gives a nontrivial bound away from the trivial maximum, not a vanishing four–point correlation. It therefore cannot be substituted for the physical gate.

Third, the averaged Chowla theorem of Matomäki et al. [7] obtains cancellation after averaging the shift parameters. It applies to correlations of arbitrary fixed order and is the natural comparison for the two– and four–point terms produced after squaring. The TPC shifts, by contrast, occupy a sparse, algebraically prescribed family tied to one CRT modulus. Shift–average cancellation does not imply a uniform estimate on that family. The quantitative

savings in that method are also logarithmic, not the K^{-1} -scale saving demanded by the endpoint ledger.

Local Fourier results change a different quantifier. The short-interval Fourier-uniformity theorem of Matomäki et al. [8] averages the origin of the interval while taking a supremum over linear phases. Its higher-order extension proves averaged local nilsequence and Gowers uniformity [9]. These are among the strongest available tools for phase-uniform cancellation on short intervals, but a contiguous interval with averaged origin is not a length- K progression of step M at one specified large modulus. Neither paper contains the row-synthesis norm required here.

For fixed shifts there are two further landmarks. Tao [11] proves the two-point Chowla–Elliott correlation with logarithmic averaging. Tao and Teräväinen [12] obtain unweighted two-point Chowla cancellation at almost all scales. In both comparisons the affine data are fixed before the asymptotic limit; the TPC data grow with X , and the desired bound must hold along the prescribed endpoint sequence with quantitative uniformity. A fixed-parameter or almost-all-scales theorem cannot be promoted to that moving-parameter assertion.

Finally, classical uniform exponential-sum cancellation for one Möbius factor [2] and strong Möbius orthogonality to fixed nilsequences [3] justify viewing a phase twist itself as benign. They do not survive the logical operation needed here: after the physical norm is squared, the coefficient is a product of two or four Möbius values rather than a single Möbius function against a deterministic phase.

10.3 The precise remaining upgrade

The missing input is therefore not another conditioning theorem for the finite Fourier bank. It is a moving-parameter decorrelation estimate, uniform over the prescribed CRT modulus $M = M(X)$, the minimal phases $0 \leq \nu < q = q(X)$, the physical row family, and the terminal support. It must control a natural L^2 synthesis containing both two- and four-Möbius off-diagonal terms at the diagonal scale. Existing results instead average shifts, interval origins, moduli, or scales; use logarithmic averages; keep the affine data fixed; or treat only one multiplicative factor.

Accordingly, none of the cited theorems proves the twist-stable physical estimate, a parity breach, a Hardy–Littlewood asymptotic, or the twin-prime conjecture. Their correct role is to identify the nearest available inputs and, just as importantly, to expose the exact quantifier upgrade that a positive continuation would have to supply.

11 The corrected route after phase-geometry closure

Within the declared class of unimodular banks, positive weights, universal recovery, and positive square-energy estimation, the finite question is now settled. Exact recovery and row decorrelation satisfy a sharp conservation law; enlarging a bank inside this class cannot change it. The remaining options must use information absent from an arbitrary Hilbert column family or leave this class.

11.1 What is unconditional

This paper proves without a Möbius hypothesis:

- (i) the weighted Loewner domination for every exact recovery bank;
- (ii) the optimal pseudoinverse formula for recovery amplification;
- (iii) the sharp product law between recovery amplification and coherent row eigenvalue;
- (iv) rigidity at unit amplification and the exact cost of complete row orthogonality;

- (v) all integer points of the sharp boundary, and its continuous weighted extension, realized by Fourier banks;
- (vi) stability under L^2 -approximate recovery;
- (vii) diagonal invariance for Hilbert row atoms;
- (viii) tensor conservation for the minimal TPC-39 alias bank; and
- (ix) equality-slice rigidity for arbitrary coupled row-alias phases.

These statements close the coefficient-blind positive phase-frame route. They do not estimate the physical coefficient vector.

11.2 Route C: coefficient-specific coherent-mode avoidance

Let G be a row Gram arising from a convenient redundant phase bank, and let v_{phys} denote the actual vector of row-Möbius coefficients after all masks and reassembly. A reusable positive theorem would bound

$$\|\Pi_{\text{coh}} v_{\text{phys}}\|^2 \tag{108}$$

at a scale smaller than the operator worst case, where Π_{coh} is the high-eigenvalue row subspace. The present no-go does not address such a coefficient-specific projection. A proof would have to exploit Möbius/source signs, the row-gcd mask, or a dispersion identity that is false for arbitrary atoms.

11.3 Route S: a signed amplitude estimate

The Loewner obstruction applies when the analytic step replaces the exact recovery combination by a positive square energy. One may instead retain the signed or complex recovery functional before Cauchy. At the level of the original scalar shell, this means estimating the matched bilinear polarization directly rather than controlling both column energies separately. Such a route can use cancellation between polarizations and between phase measurements, but no coefficient-free frame inequality supplies that cancellation.

The required theorem must be stated for the actual physical functional. Calling it a consequence of Fourier orthogonality would merely reinsert the target under another name.

11.4 Route M: the explicit moving Möbius correlation

At the terminal layer, the remaining same-row term contains

$$\mu(n)\mu(n + dM) \tag{109}$$

for structured, scale-dependent d , while cross-row expansion contains an affine four-Möbius form with the physical row weights. An averaged theorem for a declared subfamily of rows, aliases, auxiliary triples, or shifts would be genuine partial progress if it produces a quantitative off-diagonal saving beyond the invariant atomic diagonal.

Nearby results average at least one different variable, use logarithmic weights, keep the affine data fixed, or obtain only logarithmic savings. They do not currently give the $X^{-1/400}$ -scale improvement needed by the endpoint packet. The next arithmetic paper should therefore target a specific sublayer and record its exact quantifiers rather than invoke a generic “Möbius randomness” principle.

11.5 Route W: a new superwindow architecture

One may abandon the orbit–prime coordinate constraints and construct a composite no–wrap modulus closer to qM . This would require new proofs of zero–face absorption, unit coordinate ranges, target synthesis, and physical completion. No–wrap alone does not imply suppression of the equality coefficient: that coefficient and its synthesis norm must be recomputed in the new architecture. If they remain unsuppressed, as they do in the fixed orbit–prime system of [section 9](#), a physical sign estimate still remains.

11.6 Parity boundary

The two– and four–Möbius terms are consistent with the classical parity obstruction for sieve methods [4, 5]. Recovery–decorrelation uncertainty neither proves that these correlations are large nor rules out a coefficient–specific cancellation theorem. It proves only that universal positive phase geometry cannot be that theorem. No parity breach, Hardy–Littlewood asymptotic, or twin–prime conclusion follows.

12 Exact certificate and claim ledger

The structural proofs above are exact. A companion standard–library certificate checks their finite cyclic realizations using only integers, `Fraction` arithmetic, modular character orthogonality, and exact prime–order cyclotomic coordinates. It does not use floating–point eigenvalues or numerical tolerances.

12.1 Certified identities

The frozen certificate verifies:

- (i) selected R -point DFT banks containing the constant frequency, including the exact recovery amplification $A = r$, rank r , spectrum R/r , and product equality $A\lambda = R$;
- (ii) balanced and unbalanced short–cycle residue banks, their bucket spectra, and equality in the product bound exactly for equal bucket sizes;
- (iii) all folded collisions and the spectrum $2, 1, 0$, with multiplicities $L, 1, L$, for the minimal alias bank through the frozen range;
- (iv) the row–alias tensor spectrum

$$2R/r \quad (rL \text{ times}), \quad R/r \quad (r \text{ times}), \quad (110)$$

together with amplification conservation;

- (v) exact Loewner Gram factorizations for deterministic root–of–unity phase banks; and
- (vi) the endpoint exponent identity

$$\frac{267}{400} - \frac{1}{400} = \frac{266}{400} = \frac{133}{200}. \quad (111)$$

Running

```
python experiments/tpc40_certificate.py
python -O experiments/tpc40_certificate.py
```

produces byte-identical canonical JSON records. The frozen run passes

$$\boxed{2,230,446 \text{ exact checks.}} \quad (112)$$

It includes 2,748 selected DFT banks, 4,081 unbalanced bucket compositions, 19,656 row-alias tensor cases, and 300 exact cyclotomic Loewner cases. Its certificate digest is

d925148e93bc0ecd57a3958bc6ffc0447f5116cfb0f188844e8b1c93b96020a0.

The SHA-256 hashes of the canonical JSON record and source are, respectively,

94e68bc4884d80fcfc685edbee34f851c097bd49d425d31c47cd2b76ed751bd7,
2e15f17ced216acc300163e543d73d7b14878127b31478410983ff88aed6e2f3.

12.2 What the certificate does not certify

The computation verifies finite identities and implementation fidelity. It is not statistical evidence for random Möbius signs and not a test of an asymptotic physical estimate. The JSON claim ledger explicitly marks as false any assertion that the certificate proves a twist-stable physical bound, two- or four-Möbius cancellation, a parity breach, a prime-pair asymptotic, or infinitely many twin primes.

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